SMOOTH COUNTEREXAMPLES TO THE C^∞ LOEWNER AND CARATHÉODORY UMBILIC-INDEX CONJECTURES

ARGUS OPEN RESEARCH PROJECT

Abstract. Problem 8.2 in Ghomi's problem list on geometry of curves and surfaces asks to show that the index of any isolated singularity of a principal line field on a surface is at most 1 (Loewner's conjecture). While Hamburger proved the real-analytic case, we document and verify the C^∞ counterexample family g_k of Alpöge (2026) and its Lean 4 formalization, which constructs a C^∞ surface germ at the origin with an isolated umbilic of index $1 + k/2 = 3/2 > 1$ at $k = 1$ (and a smooth convex body with a single umbilic of index 2 at $k = 2$).

1. Summary of the Verified Formal Theorem

Theorem 1.1 (Alpöge, 2026; Lean 4 verified)**.** *In the C^∞ category:*

(1) *For every integer $k \geq 1$, there exists a C^∞ function $g_k : \mathbb{R}^2 \to \mathbb{R}$ whose trace-free Hessian $\left(\partial_{xx} g_k - \partial_{yy} g_k,\ 2\partial_{xy} g_k\right)$ has an isolated zero at the origin with winding number $2 + k$, corresponding to a principal-line umbilic index of $1 + k/2 > 1$.*

(2) *Taking $k = 1$ gives an isolated C^∞ umbilic of index $3/2 > 1$, refuting the C^∞ Loewner conjecture.*

(3) *Taking $k = 2$ and extending to S^2 yields the support function of a C^∞ strictly convex body in $\mathbb{R}^3$ with a unique umbilic point (of index 2), refuting the C^∞ Carathéodory conjecture.*

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